



CONTACT

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EDUCATION

- FRIEDRICH ALEXANDER UNIVERSITY**
Erlangen, Germany
 - Master of Science in Physics (Astrophysics and Astro-Particle Physics)
 - 2022 - Now
- MUMBAI UNIVERSITY**
Mumbai, India
 - Bachelor of Science in Physics
 - 2018 - 2021

SKILLS

- Python / Jupyter / Linux
- ML / Transformer Models / VLA
- NVIDIA Isaac Sim / Omniverse
- Robotic Manipulation / ALOHA
- Copilot Agents / AI Automation
- PyVISA / Data Acquisition
- Scikit-learn / XGBoost / LightGBM
- GPyTorch / MLflow
- Optical Systems / PMT Testing
- Power BI / Superset / Looker
- Autodesk Fusion / 3D Printing
- Inkscape / Canva / Photoshop
- Technical Feasibility Analysis
- Technical Writing / Leadership

LANGUAGES

- English (C1)
- German (~B1)
- Marathi (Native)
- Hindi (Native)

SAUJANYA DUBALE

PROFILE

linkedin.com/in/saujanya-dubale

I am a constant learner fuelled by curiosity. I think in first principles: I observe, analyse, identify problems, debug, and conclude. I seek out meaningful problems and create solutions that deliver value and drive positive change.

WORK EXPERIENCE

Siemens Healthineers

Working Student | Embodied AI & Robotic Manipulation R&D

04/2026 - Present
Forchheim, Germany

- Evaluating Vision-Language-Action approaches for robotic manipulation in industrial settings, focusing on technical feasibility, simulation-to-prototype workflows, and real-world engineering constraints.
- Investigating VLA-based robotic-arm workflows for perception-guided manipulation, action generation, and embodied AI task execution.
- Exploring robotics simulation and development environments including NVIDIA Isaac Sim, NVIDIA Omniverse, Isaac robotics stack, and ALOHA-based manipulation frameworks.
- Working with Linux-based development environments, open-source robotics repositories, and transformer-based model setups for experimentation and technical assessment.
- Building proof-of-concept workflows connecting simulation, model inference, and prototype deployment for lightweight hardware-oriented robotic setups.
- Comparing alternative robotics AI pipelines based on integration effort, scalability, industrial relevance, and practical deployment feasibility.

Siemens Healthineers

Working Student | AI Automation & Knowledge Management

03/2025 - 04/2026
Erlangen, Germany

- Built multiple Copilot agents for document search and Q&A, improving discoverability and reuse of internal knowledge.
- Developed an interview-driven support agent to address recurring client/customer questions, with multilingual (English, Chinese, German) audio/video transcription and response generation.
- Performed data cleaning and knowledge-base hygiene: filtered large internal file collections to identify human-relevant content, remove noise/duplicates, and improve search quality.
- Supported data analysis and reporting: cleaned/validated datasets, created recurring summaries, and prepared insights for internal updates and decision support.
- Owned editorial workflow for the CBL Connect newsletter: sourced topics, interviewed stakeholders/team leads, wrote/edited articles, and coordinated reviews to meet publishing timelines.
- Produced internal comms content (announcements, summaries, short updates) to translate team inputs into clear, employee-friendly messaging.
- Designed and maintained an interactive internal webpage for Task-Based Services (intranet) to organize key resources/updates and improve discoverability for employees.

Erlangen Centre Of Astroparticle Physics (ECAP) (FAU)

Hilfs-Wissenschaftler and Student Researcher (Thesis)

06/2024 - Present
Erlangen, Germany

- Built a complete detector-characterization pipeline to evaluate ANTARES photomultiplier tubes for sustainable reuse in SWGO.
- Designed and commissioned a light-tight quantum-efficiency measurement setup integrating a xenon lamp, monochromator, picoammeter, optical alignment system, motorized stages, and custom 3D-printed components.
- Developed Python/PyVISA automation for instrument control, wavelength scans, current readout, metadata logging, and reproducible data acquisition.
- Performed photodiode validation tests, optical alignment, calibration checks, and systematic debugging of hardware, connectivity, and measurement-stability issues.
- Extracted and prepared 150+ ANTARES photomultiplier tubes from optical modules for laboratory characterization.
- Analyzed wavelength-dependent QE data to assess sensor aging, identify measurement artifacts, and classify PMTs for potential SWGO reuse.
- Applied machine-learning methods for anomaly detection, predictive modeling, and quality-control support in PMT performance analysis.

WORK EXPERIENCE

Promeos GmbH

Working Student | AI R&D (SMEs) and Office Associate

09/2023 - 03/2024

Nürnberg, Germany

- Assisted the CEO with research on practical AI implementation opportunities for small and medium-sized enterprises.
- Prepared cost analyses for AI adoption, comparing potential tools, implementation effort, expected value, and operational feasibility.
- Designed marketing and business communication materials, including brochures, logos, and visual graphics for company presentations and outreach.

Board of European Students of Technology

Grants Department | Data and Business Analysis Department

01/2025 - 02/2026

Europe

- Grants Co-ordinator for Advocacy Team Project (ATP) : Primary contact point between Grants Department and ATP team
- To study and write the ATP grant application that BEST submitted to European Youth Foundation
- Responsible for keeping track of what we promised in the application and communicating the needs to the Local BEST Group and the team.
- Designed and delivered sessions and supported Europe-wide dissemination of program results through in-person delivery and partner outreach
- Co-created an advocacy toolkit (templates, guidance, resources) during a multi-day intensive, enabling local branches across Europe to integrate advocacy into standard operations.
- Restructuring and updating of internal knowledge management platform for Data and Business Analysis Department
- Creation of guidelines for uniform use of Google Workspace for documentation and Apache Superset and Google Looker for visualization

3foldtec

Administrative & Operations Assistant

11/2021 - 08/2022

Thane, India

- Managed production inventory, keeping an overview of production and assimilating/negotiating supplies.
- Managed top 10 clients relations with understanding their needs and finding new ones.
- Handling demands and requirements of clients based on market needs.

High Serve Society & Knowter - The Knowledge Barter

Founder

03/2020 - 08/2024

Mumbai, India

- High Serve Official was a sports management startup dedicated to delivering unparalleled experiences to hardworking and passionate players. Gave over 2000+ athletes a starting point to showcase their talent.
- Knowter was a skill/knowledge-sharing platform built around relatability, accessibility, and user freedom.
- Built an MVP matchmaking platform connecting people via freely available resources; onboarded 80+ early users.
- Formerly Scienspiration: an outreach initiative bridging underprivileged and privileged students through context-based science learning. Reached 1,000+ students via schools/colleges, delivering interactive sessions and science learning activities.

CERIFICATIONS

Application of AI in Healthcare

University of Lisbon

Sustainable Transport & Green Logistics

National Technical University of Athens

Next-Generation Materials & Advanced Manufacturing

University of Ljubljana

Silicon Thin Film Solar Cells

Ecole Polytechnique

Density Functional Theory

Ecole Polytechnique

ORGANIZATIONS

Board of European Students of Technology (Erlangen) e.V

Full Member

Full member of BEST Erlangen representing FAU across Europe; vote/propose governance changes, contribute to international teams (Grants, Advocacy, Data/Business Analysis), and design & deliver sessions.

Space Gene, FAU

Vice President

Lead teams and workflows, drive event ideation and execution, liaise with professors and sponsors, run meetings, and represent the organization externally.

Leo Club of Vashi Gold (NGO)

Member (Service Volunteer)

Planned and delivered community service initiatives end-to-end, leading multiple events and coordinating volunteers to drive measurable local impact.